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$$\begin{aligned} & \int \frac{dx}{\sqrt{x^2 + x + 6}} \\ &= \int \frac{dx}{\sqrt{(x^2 + x + \frac{1}{4}) - \frac{1}{4} + 6}} \\ &= \int \frac{dx}{\sqrt{(x + \frac{1}{2})^2 + \frac{23}{4}}} \\ & t = x + \frac{1}{2} \Rightarrow dt = dx \\ &= \int \frac{dt}{\sqrt{t^2 + \frac{23}{4}}} \\ &= \ln \left| t + \sqrt{t^2 + \frac{23}{4}} \right| + C \\ &= \ln \left| x + \frac{1}{2} + \sqrt{(x + \frac{1}{2})^2 + \frac{23}{4}} \right| + C \\ &= \ln \left| x + \frac{1}{2} + \sqrt{x^2 + x + 6} \right| + C \end{aligned}$$

References